

Product & Technical Specification

Interactive "What-If" Dynamic Scenario Simulator

Version: 1.0

Target Audience: Engineering Leads, Enterprise Architects, Solutions Architects, CTOs

Tech Stack Foundations: LangGraph, LLM Orchestration, React / Canvas Visualization, Graph Engine

1. Executive Summary & Vision

1.1 Problem Statement

Architectural decisions are frequently evaluated using static Trade-off Matrices (e.g., "Cloud Native vs. Hybrid Refactoring"). While these matrices capture high-level qualitative pros and cons, they fail to model **execution reality**: temporal cost spikes, team bandwidth bottlenecks, sequence dependencies, and cumulative operational risks over time. Consequently, architecturally sound plans often fail during delivery due to unpredicted execution constraints.

1.2 Solution Vision

The **Interactive "What-If" Dynamic Scenario Simulator** bridges the gap between static architecture design and real-world execution dynamics. By linking directly to uploaded Architecture Decision Records (ADRs), system design blueprints, and enterprise tech stack documents, the simulator uses a **LangGraph-driven multi-agent framework** to parse architectural topology, map transformation pathways, and simulate real-time execution outcomes under dynamic variable adjustments (e.g., budget limits, hiring freezes, aggressive timelines, or legacy system sunset goals).

2. Core User Persona & Primary Use Cases

2.1 Target Persona

- **Enterprise / Solutions Architects:** Needs to validate architectural pathways against financial and operational realities before presenting to executive leadership.
- **VP of Engineering / CTO:** Needs visibility into implementation risks, temporal cost curves, and resource allocation across competing initiatives.

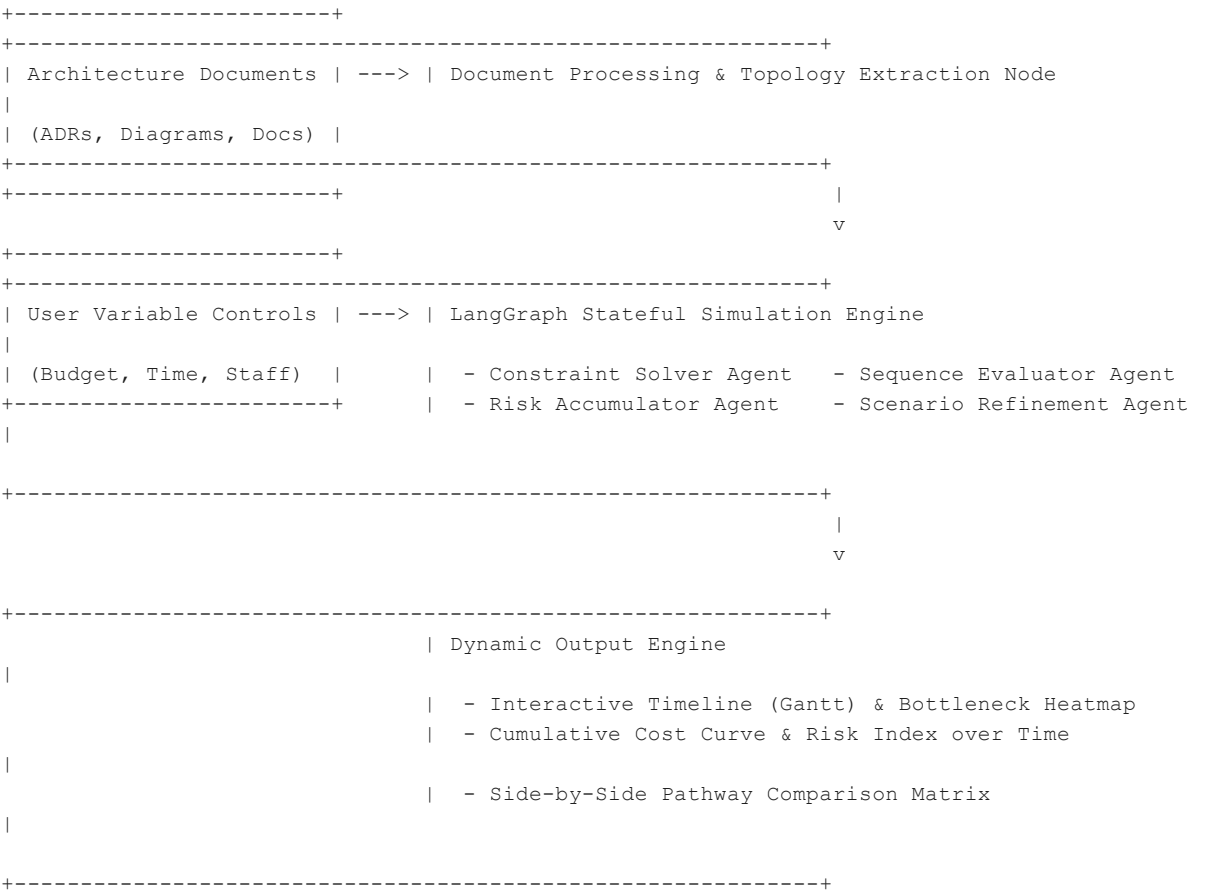
2.2 Key Use Cases

1. **Pathway Comparison:** Compare "Full Cloud Migration" vs. "Hybrid Refactoring" vs. "Strangler Fig Modernization" under identical resource and time constraints.
2. **Constraint Stress Testing:** "What happens to our risk index and delivery date if we cut the cloud migration budget by 25% or lose 2 senior DevOps engineers in Q2?"

3. **Dependency & Sequence Optimization:** Identify critical path bottlenecks and circular execution risks before engineering resources are committed.

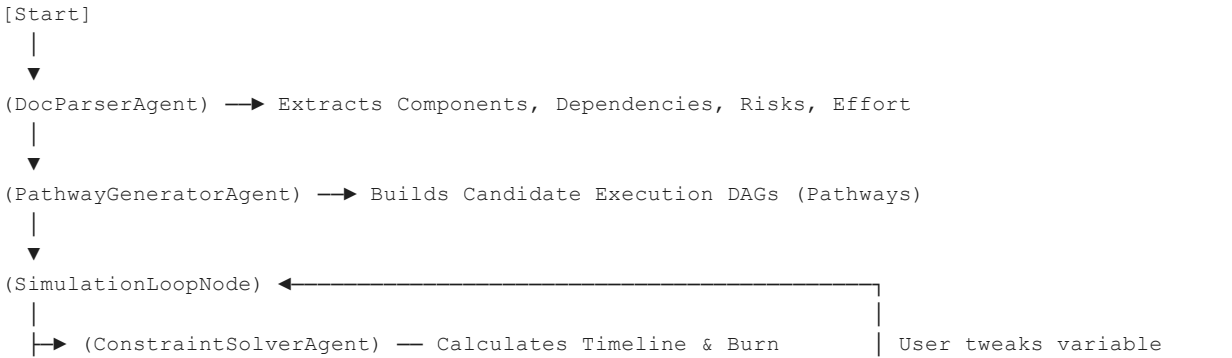
3. Key Architecture & System Design

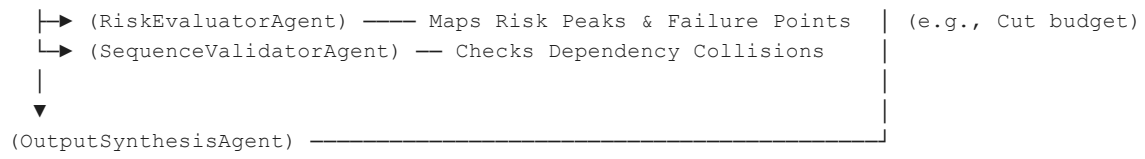
3.1 High-Level Architecture



3.2 LangGraph Workflow & State Graph

The simulation is modeled as a stateful, iterative multi-agent graph managed by **LangGraph**.





4. Functional Requirements

4.1 Architecture Document Ingestion & Context Parsing

- **Input Formats:** PDF, Markdown, Confluence Exports, OpenAPI/AsyncAPI, Architecture Decision Records (ADRs).
- **Extraction Engine (LLM Node):**
 - Parses architectural components, infrastructure dependencies, data flows, and legacy integrations.
 - Estimates task complexity, required skill sets (DevOps, Data Eng, Backend), operational risk weighting, and decommissioning dependencies.
 - Allows manual user override/tuning of parsed architectural entities.

4.2 Dynamic Variable Control Panel

Users can interactively manipulate sliders and toggles to dynamically recalculate scenarios in real time:

1. **Financial Constraints:**
 - Total Capital Expenditure (CapEx) Cap
 - Maximum Monthly OpEx Burn Rate
2. **Time & Schedule Constraints:**
 - Hard Delivery Deadline (e.g., Datacenter Contract Expiration Date)
 - Legacy System Sunset Speed Target
3. **Resource & Bandwidth Constraints:**
 - Full-Time Equivalent (FTE) Allocation by Role (DevOps, Cloud Architects, Security, QA)
 - Onboarding Velocity (ramp-up delay for contractors/new hires)
4. **Risk & Governance Thresholds:**
 - Maximum acceptable concurrency of high-risk operational migrations
 - Operational downtime tolerance (Zero-downtime vs. Maintenance Window)

4.3 Simulation & "What-If" Analytics Engine

- **Temporal Cost & Burn Simulation:** Computes capital consumption over time, alerting when monthly spend exceeds allocated budgets.
- **Cumulative Risk Profile Mapping:** Calculates compound risk based on simultaneous high-risk changes (e.g., migrating database schema while changing identity providers simultaneously).
- **Sequence & Bottleneck Identifier:** Identifies resource starvation (e.g., "DevOps team is overloaded in Month 3 by 240%, delaying Phase 2 by 8 weeks").
- **Pathways Comparative Execution:** Side-by-side execution trace contrasting Path A (e.g., Rehost) vs. Path B (e.g., Refactor) vs. Path C (e.g., Replatform).

4.4 Interactive Conversational Copilot

- Embedded AI Assistant backed by LangGraph state history.
- **Supported Commands / Prompting Capabilities:**
 - "What if we delay the database migration to Q3 due to a hiring freeze?"
 - "Suggest an optimized sequence that keeps our monthly spend under \$50k while meeting the December datacenter exit deadline."
 - "Show me the highest-risk phase in the Hybrid Refactoring pathway."

5. Visual UI & User Experience Specifications

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|  NAVBAR: Architecture Project: E-Commerce Cloud Transformation | [Export Plan] [Save
Scenario] |
+-----+-----+-----+
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|  PARAMETER CONTROLS (WHAT-IF)          | | PATHWAY COMPARISON & TIMELINE VISUALIZATION
|
|                                     | |
|
| Active Pathway: [ Hybrid Refactor v] | | [ Pathway A: Rehost ]   [ Pathway B: Hybrid Refactor
] |
|                                     | |
|
| Budget Cap: $1.2M                    | | TIMELINE & BOTTLENECK GANTT CHART
|
| [=====|-----]                   | | Month 1   Month 2   Month 3   Month 4   Month 5
|
|                                     | | [DB Migration ==>][ Auth Redesign !!! ]
|
| Max Monthly Burn: $150k              | |                                     [ Cloud Infrastructure ==>]
|
| [=====|-----]                   | | ⚠ Bottleneck: DevOps over-allocated (+180%) in Month
3 |
|                                     |
+-----+-----+-----+
|  Staff Allocation (FTEs):              | | SIMULATION ANALYTICS DASHBOARD
|
| - DevOps: [ 3 ]                        | |
|
| - Backend: [ 5 ]                       | | CUMULATIVE COST ($)          OPERATIONAL RISK INDEX
|
| - Security: [ 1 ]                      | | |      /--- (Path B)          |      /\--- (Path A)
|
|                                     | | |  --/                          |      /  \
|
| Timeline Limit: 12 Months              | | |  /----- (Path A)          |  ---/    \--- (Path B)
|
| Risk Tolerance: [ Medium v ]           | |  +-----                    +-----
|
+-----+-----+-----+
+-----+-----+-----+
|  AI SIMULATION COPILOT
|
```

| Chat: "What if we double security FTEs during Month 2?"

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6. Detailed Multi-Agent Prompting Strategy

6.1 Constraint Solver Agent Prompt

SYSTEM PROMPT:

You are an expert Enterprise Delivery & Architecture Simulation Agent.

Your job is to schedule architectural transformation tasks while honoring strict resource, financial, and temporal constraints.

INPUTS:

1. Architectural Task Graph (Tasks, dependencies, estimated person-weeks, required roles).
2. User Constraints (Monthly budget cap, FTE counts by role, timeline limits).

RULES:

- A task cannot start until all its dependencies are 100% complete.
- Total required hours for concurrent tasks cannot exceed available FTE bandwidth for any role.
- If bandwidth is exceeded, queue tasks based on critical path priority.
- Track monthly cost burn rate based on active task staffing.

OUTPUT FORMAT:

Return a JSON structure containing:

- Monthly execution schedule.
- Identified resource bottlenecks (Role, Month, Over-allocation percentage).
- Overall feasibility flag and predicted completion date.

6.2 Risk Accumulator Agent Prompt

SYSTEM PROMPT:

You are an Enterprise Risk Assessment Agent for Cloud Infrastructure Transformations.

You evaluate the concurrent execution of architectural migration tasks to detect compounded operational risk.

INPUTS:

- Scheduled execution timeline from Constraint Solver Agent.
- Task Risk Factors (0.0 to 1.0) and architectural domain contexts (Data, Identity, Network, App Logic).

RISK CALCULATION RULES:

- High-risk tasks running in parallel multiplier: If two tasks with risk > 0.6 run concurrently in interconnected domains, compound risk increases exponentially ($Risk_{total} = 1 - \prod (1 - Risk_i) \times 1.3$).
- Flag any month where total operational risk index exceeds the defined Risk Tolerance Threshold.

OUTPUT FORMAT:

JSON containing month-by-month risk indices, identified critical risk windows, and specific mitigation recommendations (e.g., "Stagger DB migration and Identity Provider swap").

7. Technical Requirements & Performance KPIs

Metric	Target Standard	SLA / Requirement
Document Parsing Latency	< 15 seconds	Parsing typical 20-page ADR / System Spec
Simulation Recalculation Time	< 1.5 seconds	Interactive slider adjustment response
Graph Constraint Precision	100% Deterministic	Dependency ordering must never violate DAG constraints
Export Formats	PDF, JSON, MS Project XML, Mermaid.js	Exportable execution plans for stakeholders

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